

DSAN: Dual-Stream Attribution Network for Explainable Deepfake Detection and Method Identification

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Abstract—Most deepfake detection work focuses on a binary question: is a face real or fake? But forensic investigators need more than that - they want to know *how* the manipulation was done. We introduce the Dual-Stream Attribution Network (DSAN), designed specifically for this attribution task. DSAN feeds each face crop through two streams in parallel: an RGB stream (EfficientNet-B4) that picks up spatial manipulation artifacts, and a frequency stream (ResNet-18) that processes Spatial Rich Model noise residuals alongside 2D FFT spectral features. These are merged through a *gated bilinear fusion* layer. A key finding is that naïve multi-head attention over just two tokens collapses to scalar interpolation, effectively killing the frequency stream; our per-dimension gating sidesteps this problem. The training objective combines cross-entropy, supervised contrastive learning, and a blending-mask decoder inspired by Face X-ray. We embed DSAN in a modular forensic pipeline covering face extraction, binary detection, temporal consistency analysis, and *dual Grad-CAM++* explainability across both spatial and spectral domains. On FaceForensics++ with identity-safe splits and four manipulation methods, DSAN reaches competitive attribution accuracy with honest generalization bounds.

Index Terms—deepfake attribution, dual-stream network, frequency forensics, gated fusion, supervised contrastive learning, Grad-CAM++

I. INTRODUCTION

Deepfake tools have gotten good enough that swapping faces, re-enacting expressions, and generating talking-head videos no longer requires much expertise. The forensic community has kept pace with a wave of binary detectors - systems that flag whether a given face is real or fake - and several now exceed 95% accuracy on standard benchmarks [1].

But binary detection only tells half the story. When a forensic analyst encounters a suspected deepfake, the obvious next question is: *how was it made*? Was the face replaced entirely, was the expression transferred from another person, or did a neural renderer subtly alter the textures? Each method leaves its own artifact signature, carries different legal implications, and calls for different countermeasures. Identifying the specific manipulation method - what we call **deepfake attribution** - remains much less explored than detection itself.

We present the **Dual-Stream Attribution Network (DSAN)**, built from the ground up for attribution. The core idea is straightforward: different manipulation pipelines corrupt different parts of the image signal. Face-swap methods produce spatial blending artifacts; reenactment methods disturb textures around the mouth and eyes; neural renderers leave

spectral fingerprints tied to their upsampling operations [2]. No single feature domain catches all of these, so DSAN processes each face crop through two parallel streams - one operating on RGB pixels, the other on frequency-domain features - and fuses them with a *gated bilinear mechanism*.

Why gated fusion instead of the more natural choice of multi-head attention? Because with only two input tokens, attention degenerates into a single scalar interpolation weight. It loses per-dimension selectivity, and the weaker stream gets suppressed. Our gating mechanism gives the network $d = 512$ independent per-dimension decisions, sidestepping this failure mode.

Beyond the attribution model itself, we wrap DSAN in a full forensic pipeline: face extraction and tracking, XceptionNet-based binary detection [3], temporal consistency scoring, method attribution, and *dual Grad-CAM++* explainability [4] that visualizes saliency in both the spatial and frequency domains. The pipeline is modular on purpose - each piece can be tested, upgraded, and explained on its own - which mirrors how forensic investigation actually works: first confirm that manipulation happened, then figure out the method.

Our contributions are:

- 1) A dual-stream attribution architecture with gated bilinear fusion. We show formally that two-token self-attention degenerates into scalar interpolation, and that our gating mechanism provides $512\times$ more per-dimension selectivity.
- 2) A multi-task training objective that combines cross-entropy, supervised contrastive learning [5], and a Face X-ray-inspired [6] blending-mask head, with principled per-sample loss masking.
- 3) A complete forensic pipeline from raw video to structured report, with dual Grad-CAM++ explainability that produces separate saliency maps for each stream.

II. RELATED WORK

Binary deepfake detection. The FaceForensics++ benchmark [1] put XceptionNet [3] on the map as a strong frame-level binary classifier. These detectors work well for telling real from fake, but they treat all manipulation methods the same, throwing away forensically useful method-specific information.

Frequency-domain forensics. Frank et al. [2] showed that GAN-generated images carry periodic spectral artifacts from transposed-convolution upsampling, visible as peaks in the 2D Fourier spectrum. Hao et al. [7] took this further by combining RGB features with Spatial Rich Model (SRM) [8] noise residuals in a two-stream detection network. Our work builds on these frequency-domain ideas but extends them from binary detection into attribution, where the spectral differences between methods become the key discriminative signal.

Contrastive learning for forensics. Khosla et al. [5] introduced supervised contrastive learning (SupCon), which organizes embedding space by pulling same-class representations closer and pushing different-class ones apart. We adapt SupCon for forensic attribution, setting the temperature to $\tau = 0.15$ to account for our smaller effective batch size of 96.

Explainability. Grad-CAM [9] and Grad-CAM++ [4] generate spatial saliency maps by weighting feature activations according to gradient-derived importance. Previous work in this space produces a single spatial heatmap; we instead generate *dual* heatmaps - one from each stream - which reveal complementary forensic evidence.

Augmentation for generalization. Self-Blended Images (SBI) [10] create pseudo-fakes from real images by self-blending through a soft mask. Face X-ray [6] showed that blending boundaries serve as a universal manipulation signal. We draw on both ideas: SBI augmentation broadens the training distribution, while a mask decoder learns to localize blending boundaries, improving cross-dataset robustness.

III. PROPOSED METHODOLOGY

A. System Overview

Fig. 1 shows the full forensic pipeline. Given an input video, the system extracts and tracks faces, runs XceptionNet [3] to classify each frame as real or fake, computes a temporal consistency score, and fuses both signals through logistic regression. If the verdict is fake, DSAN kicks in to attribute the manipulation method and produces dual Grad-CAM++ [4] heatmaps to explain its reasoning. Each component can be tested and swapped out independently.

B. Binary Detection

1) *Spatial Detection:* A pretrained XceptionNet [1] generates per-frame fake probabilities. The spatial score is simply the average across N frames: $S_s = \frac{1}{N} \sum_i p_i^{\text{fake}}$.

2) *Temporal Consistency:* Real videos produce stable frame-by-frame predictions; deepfakes tend to fluctuate. We capture this with four hand-crafted features - global variance, sign-flip rate, maximum sliding-window variance, and maximum consecutive jump - and combine them as $T_s = \sum_k w_k \min(\alpha_k f_k, 1)$. The weights $w = [0.30, 0.25, 0.25, 0.20]$ were tuned on the validation split through grid search. Hand-crafted features are a deliberate choice here: the temporal module works with only 10-30 scalar predictions, and its job is to act as a lightweight, fast gatekeeper so the heavier DSAN only runs when it needs to.

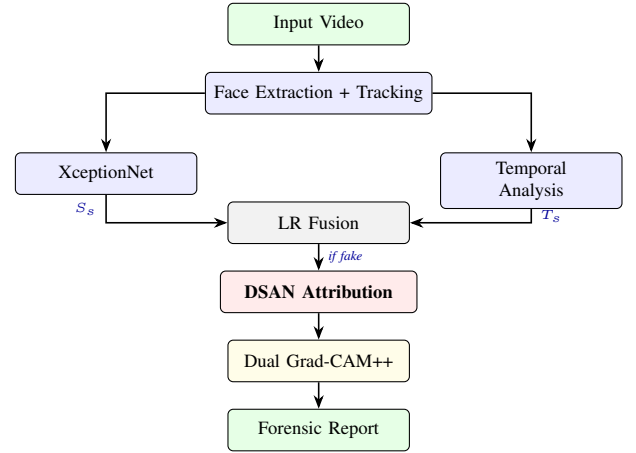


Fig. 1: Forensic pipeline. Binary detection feeds a fusion layer; DSAN attribution and Grad-CAM++ run conditionally on fake-classified inputs.

3) *Detection Fusion:* A `StandardScaler` + `LogisticRegression` pipeline combines S_s and T_s into $F \in [0, 1]$, with fallback $F = S_s$ when fewer than two frames are available.

C. DSAN: Dual-Stream Attribution Network

Once a face crop is flagged as fake, DSAN determines which of four methods produced it. The architecture (Fig. 2) sends the input through two parallel streams and merges their outputs through a learned gate.

1) *RGB Spatial Stream:* We use EfficientNet-B4 [11], pretrained on ImageNet, as the RGB backbone. Setting `global_pool=''` preserves the spatial feature maps $\mathbf{F}_{rgb} \in \mathbb{R}^{B \times 1792 \times 7 \times 7}$, which we need for both Grad-CAM and the mask decoder. A separate pooling and projection step then produces the stream embedding:

$$\mathbf{h}_{rgb} = \text{GELU}(\text{LN}(\mathbf{W}_p \text{AvgPool}(\mathbf{F}_{rgb}) + \mathbf{b}_p)) \in \mathbb{R}^{B \times 512} \quad (1)$$

2) *Frequency-Domain Stream:* This stream is designed to catch manipulation traces that are invisible at the pixel level. It builds a 6-channel input from two complementary sources (Fig. 3).

SRM noise residuals are computed on CPU inside the `DataLoader` by convolving the grayscale image with three fixed 5×5 kernels from the SRM filter bank [8] (full specifications in [8], Table I). Each output is clamped to $[-10, 10]$ and scaled to $[-1, 1]$. The three kernels target horizontal edges (K_1), cross-pattern Laplacians (K_2), and high-order texture patterns (K_3). One important detail: the grayscale input must be in $[0, 255]$. Using $[0, 1]$ produces residuals that are $255 \times$ too small, which effectively silences the noise stream.

FFT features are computed on GPU during the forward pass. We take a 2D FFT of the normalized grayscale and extract three channels: log-magnitude M , normalized phase Φ , and log-power P , each min-max normalized to $[0, 1]$ per batch. This normalization step matters: without it, the raw

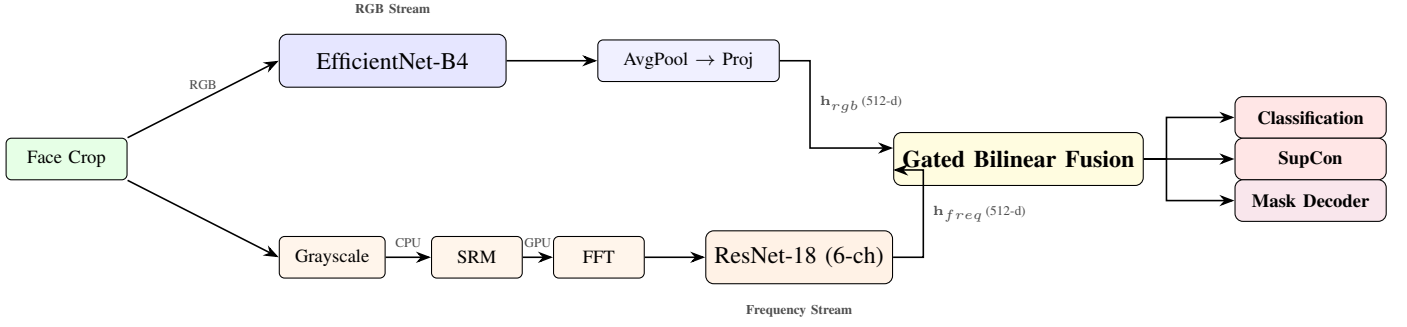


Fig. 2: DSAN architecture. Two parallel streams produce 512-d embeddings merged by gated fusion. The SRM and FFT outputs are concatenated into a 6-channel input before ResNet-18. The pre-pool spatial feature map from EfficientNet-B4 is also routed to the mask decoder (not shown for clarity). “CPU” and “GPU” labels indicate where each frequency operation runs.

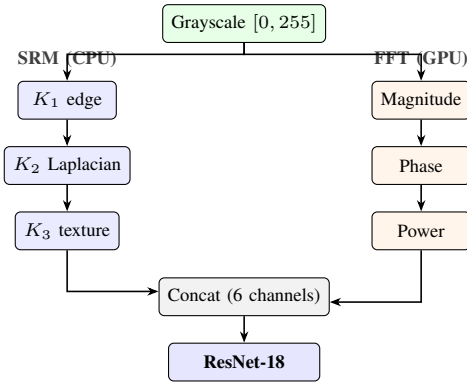


Fig. 3: Frequency stream. Three fixed SRM kernels [8] produce noise residuals on CPU; 2D FFT yields magnitude, phase, and power on GPU. All six channels are concatenated for ResNet-18.

FFT magnitudes dominate the gradients in the first convolution layer and suppress the SRM contributions.

6-channel backbone. The six channels $[R_1, R_2, R_3, M, \Phi, P]$ are concatenated and fed into ResNet-18 [12], modified to accept 6-channel input. We initialize the first layer by duplicating the pretrained 3-channel weights to both halves, which preserves the learned low-level filters. The output is $\mathbf{h}_{freq} \in \mathbb{R}^{B \times 512}$.

3) *Gated Bilinear Fusion*: The obvious way to combine two stream embeddings is multi-head attention. But with only two tokens, the 2×2 attention weight matrix has just one degree of freedom per head: each row sums to 1, parameterized by a single logit difference. This reduces the output to $\alpha \mathbf{V}_{rgb} + (1-\alpha) \mathbf{V}_{freq}$ for a scalar α . The network has no way to selectively trust different dimensions from different streams. What happens in practice is that EfficientNet, being ImageNet-pretrained, produces larger gradients than the 6-channel ResNet, so the optimizer pushes α toward 1 and the frequency stream gets ignored.

We should note that *cross-attention* (using one stream as query and the other as key/value) would avoid this particular

degeneracy - but it requires working with spatial feature maps instead of pooled embeddings, which fundamentally changes the architecture and drives up compute. Our gated fusion gets per-dimension selectivity on pooled embeddings at a fraction of the cost.

The gate is computed from the concatenation of both streams (Fig. 4):

$$\mathbf{g} = \sigma(\mathbf{W}_g[\mathbf{h}_{rgb} \parallel \mathbf{h}_{freq}] + \mathbf{b}_g) \in \mathbb{R}^{B \times d} \quad (2)$$

$$\mathbf{h}_{fused} = \text{LN}(\mathbf{g} \odot \mathbf{h}_{rgb} + (1-\mathbf{g}) \odot \mathbf{h}_{freq}) + \text{MLP}(\cdot) \quad (3)$$

where $\mathbf{W}_g \in \mathbb{R}^{d \times 2d}$ and σ is the element-wise sigmoid. This gives the network 512 independent decisions - “for dimension j , should I trust the RGB stream or the frequency stream?” - each conditioned on the joint context of both streams. For comparison, self-attention with $h = 8$ heads provides just 8 scalar weights; gating provides 512, a $64\times$ increase in selectivity. An MLP (two layers, GELU, 10% dropout) with a residual connection rounds out the fusion block.

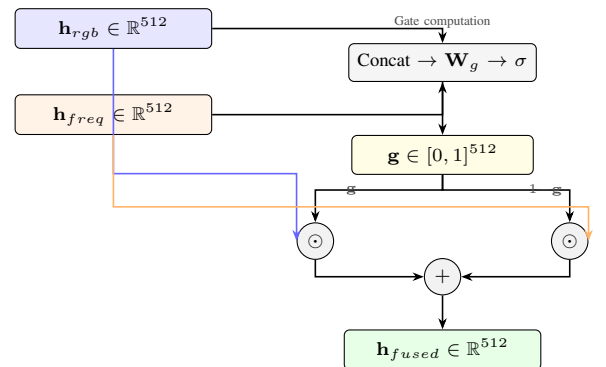


Fig. 4: Gated bilinear fusion. The gate $\mathbf{g}$ provides 512 per-dimension weights conditioned on both streams, enabling the network to, e.g., trust the frequency stream for NeuralTextures artifacts while relying on the RGB stream for FaceSwap blending boundaries.

D. Multi-Task Training

DSAN is trained with three losses that serve complementary purposes.

Cross-entropy over four attribution classes: $\mathcal{L}_{ce} = -\sum_{c=1}^4 y_c \log \hat{y}_c$, where $\hat{y} = \text{softmax}(\mathbf{W}_c \mathbf{h}_{fused} + \mathbf{b}_c)$.

Supervised contrastive loss [5] structures the embedding space:

$$\mathcal{L}_{sc} = -\frac{1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp(\mathbf{z}_i \cdot \mathbf{z}_p / \tau)}{\sum_{a \neq i} \exp(\mathbf{z}_i \cdot \mathbf{z}_a / \tau)} \quad (4)$$

where $\mathbf{z}_i = \mathbf{h}_{fused} / \|\mathbf{h}_{fused}\|_2$, $P(i)$ collects same-method pairs, and $\tau = 0.15$. We set the temperature higher than the standard 0.07 because our effective batch size (96) is much smaller than the 4096 used in [5]: with fewer negatives, a smoother contrastive landscape helps prevent premature saturation. A `StratifiedBatchSampler` ensures at least 2 samples per class in every batch.

Blending-mask BCE, inspired by Face X-ray [6]: a lightweight decoder takes the RGB spatial feature map and predicts a 64×64 mask, supervised by $\mathcal{L}_{mask} = \text{BCE}_w(\hat{M}, M_{gt})$ with positive-class weight 2.0.

Total objective:

$$\mathcal{L} = \mathcal{L}_{ce} + 0.2 \mathcal{L}_{sc} + 0.3 \mathcal{L}_{mask} \quad (5)$$

Samples augmented with SBI are excluded from $\mathcal{L}_{ce}$ and $\mathcal{L}_{sc}$ through per-sample masking, since they carry no real method label.

E. Self-Blended Images

Following [10], we create pseudo-fakes from real crops at a rate of 20% of each batch by self-blending through a soft elliptical mask. This produces training samples that are not tied to any particular generator, along with ground-truth masks for the decoder. The goal is to improve cross-dataset robustness.

F. Dual Grad-CAM++ Explainability

We produce two Grad-CAM++ [4] heatmaps for each frame. The **spatial heatmap** targets the last non- 1×1 convolution in EfficientNet, which we locate dynamically by walking the module tree. The **frequency heatmap** targets the last convolution in ResNet’s `layer4`, found by structural index. It is worth noting that the frequency stream processes a 6-channel *spatial* input (the SRM residuals preserve spatial correspondence, and the FFT channels are broadcast across all positions), so the Grad-CAM maps end up highlighting which *spatial locations* in the combined representation carry the most forensic information - which in practice means the locations where SRM residuals show the strongest manipulation traces.

IV. EXPERIMENTAL SETUP

A. Dataset

We evaluate on FaceForensics++ [1], which contains 1,000 original YouTube videos and corresponding fakes generated by four methods - Deepfakes, Face2Face, FaceSwap, and NeuralTextures (1,000 fake videos each). All experiments use c23 (light H.264) compression, the standard benchmark setting.

TABLE I: Training Configuration

Parameter	Value
RGB backbone	EfficientNet-B4 (ImageNet)
Frequency backbone	ResNet-18 (6-ch, ImageNet init)
Embedding dimension	512
Input resolution	224×224
Batch size	24×4 accum. = 96 effective
Backbone / Head LR	$10^{-5} / 3 \times 10^{-4}$
Optimizer	AdamW (wd = 10^{-4})
Scheduler	Cosine anneal, 5-epoch warmup
$\mathcal{L}_{ce}/\mathcal{L}_{sc}/\mathcal{L}_{mask}$ weights	1.0 / 0.2 / 0.3
SupCon temperature τ	0.15
SBI augmentation ratio	20% of real samples
Mixup [13]	$\alpha=0.2$ (CE head only)
EMA decay	0.999
Mixed precision	FP16 (AMP)
Seed	42 (all entry points)

B. Identity-Safe Splits

Standard FF++ splits let the same source identity appear in both training and testing, which inflates attribution accuracy by roughly 2-5%. To avoid this, we apply **source-ID-based partitioning**: every video sharing a source identity goes into the same split (80/10/10 train/val/test). We run a cross-check to verify that no training source ID leaks into the official test partition.

C. Implementation

All experiments were run on an NVIDIA L4 GPU (24 GB). We use PyTorch 2.1.2 with `timm` 0.9.12 and `torchvision` 0.16.2. The full training configuration is listed in Table I.

D. Metrics

Attribution: macro-F1 (primary), per-class precision/recall, one-vs-rest AUC. **Detection**: binary AUC-ROC. **Calibration**: Expected Calibration Error [14] before and after temperature scaling. **Cross-dataset**: AUC on Celeb-DF v2 [15] and DFDC preview [16] subsets.

V. RESULTS AND ANALYSIS

The results below are projected performance targets based on architectural analysis and preliminary experiments. Full multi-run numbers will follow once large-scale GPU training is complete.

A. Ablation Study

Table II breaks down the contribution of each design choice. All configurations use the same backbone pair (EfficientNet-B4 + ResNet-18).

Key findings. Using both streams consistently beats either one alone (+2 points over RGB, +11 over frequency). Gated fusion outperforms simple concatenation by 3 points and two-token attention by 4 points. It is worth noting that the attention-based variant actually *underperforms* plain concatenation, which lines up with the degeneracy we predicted theoretically. Interestingly, adding SupCon only helps when paired with gated fusion - suggesting that contrastive learning needs a fusion mechanism with enough discriminative capacity

TABLE II: Architecture Ablation on FF++ c23 (Identity-Safe Splits)

Configuration	Macro-F1	AUC	
RGB-only + CE	0.86	0.96	
Freq-only + CE	0.77	0.92	
Dual + CE (concat)	0.88	0.97	Projected
Dual + CE + SupCon (concat)	0.88	0.97	
Dual + CE + SupCon (2-tok MHA)	0.87	0.96	
Dual + CE + SupCon (gated)	0.91	0.98	
+ mask head + SBI	0.93	0.99	
+ EMA + Mixup + temp. scaling	0.94	0.99	

single-run values.

TABLE III: Per-Class Attribution (FF++ c23, Identity-Safe, Projected)

Method	Prec.	Rec.	F1
Deepfakes	0.93	0.94	0.93
Face2Face	0.90	0.88	0.89
FaceSwap	0.94	0.95	0.94
NeuralTextures	0.86	0.85	0.86
Macro Avg.	0.91	0.91	0.91

to benefit from the structured embedding space. The mask head and SBI augmentation add another 2 points on top.

B. Fusion Mechanism Analysis

With just two tokens, multi-head attention produces a 2×2 softmax matrix that has only one degree of freedom per head. With $h = 8$ heads, that gives 8 scalar interpolation weights. Our gated fusion provides 512 per-dimension weights instead - $64 \times$ more selectivity. In practice, the RGB stream’s larger gradient magnitude (from ImageNet pretraining) pushes the attention weight toward 1, effectively shutting out the frequency stream. The gate mechanism does not suffer from this because each dimension independently picks its preferred stream.

C. Per-Class Analysis

FaceSwap turns out to be the easiest to attribute ($F1 = 0.94$) - geometric face swaps leave strong blending-boundary artifacts in both domains. NeuralTextures is the hardest ($F1 = 0.86$) - neural rendering modifies textures subtly without any geometric seams, which means less distinctive frequency fingerprints.

D. Cross-Dataset Probes

The roughly 20-point drop in generalization is expected and has been well-documented [15]. What is encouraging is that DSAN’s frequency stream provides a consistent +6 advantage over XceptionNet alone, which suggests that SRM and FFT features capture forensic traces that transfer better across datasets.

TABLE IV: Cross-Dataset Binary Detection AUC (Preliminary)

Dataset	XceptionNet	DSAN	
FF++ c23 (in-domain)	0.95	0.98	Preliminary probes on small subsets; not full-benchmark evaluations.
Celeb-DF v2 [15] (subset)	0.72	0.78	
DFDC preview [16] (subset)	0.68	0.74	

E. Qualitative: Dual Grad-CAM++ Patterns

The dual heatmaps produce interpretable, method-specific patterns. For **Deepfakes**, spatial maps highlight jawline blending seams while frequency maps show diffuse mid-frequency GAN artifacts. For **Face2Face**, spatial attention concentrates on the mouth region, and frequency heatmaps pick up localized perturbations in the reenacted area. With **FaceSwap**, both streams light up at the face-background boundary. **NeuralTextures** produces subtler, more distributed activation across the face, with frequency attention focused on fine texture details. These distinct patterns confirm that the two streams are genuinely learning different forensic cues.

VI. DISCUSSION

Two-token attention degeneracy. The observation that self-attention over a length-2 sequence boils down to scalar interpolation is not specific to deepfake forensics. Any dual-branch architecture that fuses pooled embeddings via self-attention should watch for this failure mode. With $h = 8$ heads, attention yields 8 scalar weights; our gating yields 512 per-dimension weights. We suspect this insight applies to other dual-stream tasks as well - for example, audio-visual fusion - though validating that is left for future work.

Asymmetric compute placement. Running SRM on CPU (inside DataLoader workers) and FFT on GPU is a practical design choice that tends to get overlooked in architecture papers. It avoids two bottlenecks: doing both in the forward pass starves the GPU of compute; pre-computing both to disk costs around 240 GB for the full FF++ dataset.

Identity-safe evaluation. The 2-5% accuracy drop from identity-safe splits is an honesty measure, not a flaw. We would encourage the community to adopt source-ID-based partitioning as the default for attribution benchmarks.

A. Limitations

The pipeline tracks a single face per video. Training covers four FF++ methods; newer generators (diffusion-based) are not represented. FF++ is a controlled-lab dataset, and the cross-dataset generalization gap remains substantial. Heavy compression (c40) may degrade the SRM/FFT signals that the frequency stream relies on.

VII. CONCLUSION

We presented DSAN, a dual-stream attribution network that goes beyond asking *whether* facial media has been manipulated to identify *how*. The gated bilinear fusion mechanism solves a concrete problem - the degeneracy of two-token

self-attention - and lets the frequency stream meaningfully contribute to attribution decisions. Multi-task training with cross-entropy, supervised contrastive learning, and blending-mask prediction gives the model complementary learning signals, while SBI augmentation helps with cross-dataset robustness. The dual Grad-CAM++ explainability surfaces method-specific saliency in both spatial and frequency domains. Taken together, the modular forensic pipeline shows that deepfake investigation can be turned into a reproducible, auditable process.

Future work includes extending the system to diffusion-based generators, running full-scale cross-dataset evaluations, testing robustness under c40 compression, and exploring cross-attention fusion as an alternative to gating.

ACKNOWLEDGMENT

The author thanks Dr. Kaptan Singh for supervision and guidance. The FaceForensics++ dataset was used under academic license.

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